

Conjecture 00000003905 is false:
 $\mu((x)^{[p^e]}) = 1$ in $k[x, y]$, while $\dim R - \text{ht}(x) = 1$

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1 The conjecture

Conjecture 00000003905 (English text, verbatim): “Definition: The Frobenius power $I^{[p^e]}$ is the ideal generated by the p^e -th powers of the generators of I , and μ denotes the minimal number of generators. Conjecture: For fixed I , the number $\mu(I^{[p^e]})$ is a polynomial in p^e once $e \geq e_0$, of degree equal to the ring dimension minus the height of I , and its leading coefficient depends only on the initial ideal.” The Chinese text says the same.

Reading. R is a Noetherian ring of prime characteristic p ; since the text speaks of an initial ideal, R is a polynomial ring over a field. “A polynomial in p^e once $e \geq e_0$ of degree $\dim R - \text{ht } I$ ” means: there are e_0 and a polynomial P with $\mu(I^{[p^e]}) = P(p^e)$ for all $e \geq e_0$ and $\deg P = \dim R - \text{ht } I$. We take the coefficients of P in an arbitrary field K of characteristic 0 (for instance $\mathbb{Q}$ or $\mathbb{R}$), which covers integer and rational coefficients. We refute the degree clause, and hence the conjunction, for $R = k[x, y]$ (k any field of characteristic p , any prime p) and $I = (x)$.

Not refuted: the bare polynomiality clause (it holds here: the value is constantly 1); the leading-coefficient clause (not addressed); readings restricted to ideals with $\text{ht } I = \dim R$ (for example $\mathfrak{m}$ -primary ideals, where the predicted degree is 0); the local ring $k[x, y]_{(x, y)}$ (the same computation applies, but it is not formalized).

2 Definitions

Definition 1. For an ideal I of a commutative ring R and $q \in \mathbb{N}$, $I^{[q]}$ is the ideal generated by $\{a^q : a \in I\}$ (Lean: `frobeniusPow`). Lemma 3 shows that for $q = p^e$ in characteristic p it equals the ideal generated by the q -th powers of the elements of *any* generating set of I , so it agrees with the definition in the conjecture and does not depend on the chosen generators.

Definition 2. $\mu(J)$ is the least cardinality of a generating set of J (Mathlib: `Submodule.spanFinrank`, defined as the least cardinality of a generating set; it is set to 0 only when no finite generating set exists, which cannot happen in a Noetherian ring). $\dim R$ is the Krull dimension (Mathlib `ringKrullDim`, the supremum of lengths of chains of prime ideals) and $\text{ht } I$ is the height (Mathlib `Ideal.height`, the infimum of the heights of the minimal primes over I).

3 Proof

Lemma 3 (`frobeniusPow_eq_span_pow_gens`). *Let R have characteristic p , $q = p^e$, $S \subseteq R$. Then $(S)^{[q]} = (s^q : s \in S)$.*

Proof. $\supseteq$ is clear. For $\subseteq$ it suffices that $a^q \in (s^q : s \in S)$ for every $a \in (S)$; by induction on a : $0^q = 0$; s^q is a generator; $(a + b)^q = a^q + b^q$ in characteristic p ; and $(ra)^q = r^q a^q$. $\square$

Lemma 4 (`ringKrullDim_eq_two, height_I`). *$\dim k[x, y] = 2$ and $\text{ht}(x) = 1$.*

Proof. Mathlib’s `MvPolynomial.ringKrullDim_of_isNoetherianRing` states $\dim R[X_1, \dots, X_n] = \dim R + n$ for Noetherian R ; with $\dim k = 0$ this gives 2. Mathlib’s `Ideal.height_span_singleton_eq_one_of_mem_nonZeroDivisors` (docstring: “In a Noetherian ring, the height of a principal ideal spanned by a non-unit non-zero-divisor is one”) applies to x : it is nonzero in the domain $k[x, y]$, and it is not a unit because evaluation at 0 sends it to 0. $\square$

Lemma 5 (`mu_frobeniusPow_I`). *$\mu((x)^{[p^e]}) = 1$ for every $e \geq 0$.*

Proof. By Lemma 3, $(x)^{[p^e]} = (x^{p^e})$. A nonzero principal ideal has $\mu = 1$ (Mathlib `Submodule.spanFinrank_singleton`), and $x^{p^e} \neq 0$. $\square$

Lemma 6 (`eq_one_of_eval_pow`). *If $P \in K[t]$, $p \geq 2$ and $P(p^e) = 1$ for all $e \geq e_0$, then $P = 1$.*

Proof. The points p^e ($e \geq e_0$) are pairwise distinct in K (characteristic 0), so $P - 1$ has infinitely many roots. $\square$

Theorem 7 (`conjecture_3905_false`). *For every prime p , field k of characteristic p and field K of characteristic 0: in $R = k[x, y]$ with $I = (x)$, $\dim R = 2$, $\text{ht } I = 1$, and there are no e_0 and $P \in K[t]$ with $\mu(I^{[p^e]}) = P(p^e)$ for all $e \geq e_0$ and $\deg P + \text{ht } I = \dim R$.*

Proof. By the lemmas, any such P equals 1, so $\deg P + \text{ht } I = 0 + 1 = 1 \neq 2$. $\square$

We state the degree condition as $\deg P + \text{ht } I = \dim R$ (in Mathlib’s `WithBot  $\mathbb{N}_\infty$`), which is equivalent to $\deg P = \dim R - \text{ht } I$ because all three quantities are finite here. Here `deg` is `Polynomial.natDegree`; `frobenius_mu_polynomial_is_one` also proves $P = 1$ and `Polynomial.degree P = 0`, so the conclusion does not depend on the degree convention.

4 Lean formalization

File `lean/Conjecture3905/Basic.lean` (151 lines, `import Mathlib only`; Lean v4.33.1, Mathlib v4.33.1). Main theorems: `C3905.conjecture_3905_false`, `C3905.frobenius_mu_polynomial_is_one`; supporting: `frobeniusPow_eq_span_pow_gens`, `ringKrullDim_eq_two`, `height_I`, `frobeniusPow_I`, `mu_frobeniusPow_I`, `eq_one_of_eval_pow`. `lake build` succeeds; `#print axioms` for both main theorems reports exactly `[propext, Classical.choice, Quot.sound]`. There is no `sorry`, no `native_decide` and no added axiom.

5 Relation to the earlier submission

PR #286 (by orionsheep) used $I = (x, y)$ in $k[x, y, z]$, with $\mu = 2$ for every q . It was not merged: its main theorem was a conjunction of numeral identities such as $(2 : \mathbb{N}) = 2$ and $2 \cdot 2 - 2 = 2$, and no ring, ideal, Frobenius power or μ was defined in Lean. The reviewer asked for a version that “defines μ on ideals in Lean”. This submission defines the Frobenius power in Lean and proves that it does not depend on the generators. It uses Mathlib’s μ , Krull dimension and height, proves $\dim k[x, y] = 2$, $\text{ht}(x) = 1$ and $\mu((x)^{[p^e]}) = 1$ for every e , and proves that no interpolating polynomial has degree $\dim R - \text{ht } I$. We switched to the principal ideal (x) because $\mu = 1$ is immediate there, whereas $\mu((x^q, y^q)) = 2$ would need a separate non-principality proof.

Sources

- Mathlib, files `RingTheory/KrullDimension/Polynomial.lean`, `RingTheory/Ideal/Krull sHeightTheorem.lean` and `Algebra/Module/SpanRank.lean`.
- PR #286: <https://github.com/The-Last-Math-Competition/The-Last-Math-Competition/pull/286>.